

Landslide susceptibility modelling using deep-learning and machine-learning methods-A study from southern Western Ghats, India

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Abstract

Accurate and reliable landslide susceptibility modelling is essential for regional landslide risk assessment and mitigation. In the present study, a spatially explicit deep learning neural network (DNN) based landslide susceptibility model for Wayanad district of Kerala state in India has been developed and it is compared with conventional random forest machine-learning model. The Wayanad was severely affected by devastating landslides during 2018 extreme climatic event and absence of an accurate landslide susceptibility model caused extensive casualties. A geospatial database was developed using 263 previous landslides event locations and twelve geo-environmental variables such as lithology, soil texture, land use/ land cover, slope angle, slope aspect, topographic wetness index, distance from the road, distance from the streams, distance from the lineaments, convergence index, profile and plan curvatures were used for susceptibility modelling. The present investigation used search grid to obtain best hyper tuning parameters and training data is split for 5-fold cross validation. Subsequently the best model is chosen based on minimal log loss, RMSE and maximum area under curve values (AUC) value for final prediction. The results are validated with receiver operating characteristics curve with AUC (ROC-AUC) and confusion matrix-based parameters. Results showed that DNN outweigh RF model with 93.4% (AUC=0.934) accuracy over 91.1% in training frame. In validation section DNN obtained 88.7% accuracy while RF shows only 85%. It is also noted that DNN shows better spatial pattern in predicting the probability of landslide occurrence whereas a spatial overfitting is found in RF model. The proposed DNN model is trust worthy and can be used for future hazard mitigation and land use planning in the study area.

Keywords: Landslide, deep learning neural network, GIS, Wayanad, India

Introduction

Landslides are one of the most frequent natural hazards, subscribed by geological as well as hydrological factors, affecting the tropical mountainous regions across the world. Globally, nearly 66 million people live in highly landslide prone areas, where approximately 17% fatalities are caused by landslides (Sassa and Canuti, 2008). About 16% of the world's landslides (non-seismic) during the last decade were reported from India (Froude and Petley, 2018), the country affected by repeated landslides especially along the Himalayan belt and Western Ghats (WG). New bigdata and data integration methods such as Deep learning coupled with geospatial technology offers new opportunities for both local and regional monitoring, modelling and early warning of landslides (van Natijne et al., 2020) Aim of the present study is to model landslide susceptibility using advanced deep learning neural network and compare its performance with traditional RF model.

Study area

The present study is carried out in Wayanad district of Kerala state in India (area=2130km²) (Figure 1). Geomorphologically Wayanad is a table land, located in the southern tip of Deccan Plateau. Geologically, Wayanad is an extension of Dharwar Craton, which is one of the significant crustal blocks in the Southern Granulate Terrain (SGT). The steep scarps of the Wayanad Plateau are deeply dissected by fluvial action to form V-shaped valleys and the slopes are covered with soil matrix of varying thickness. The study area experiences monsoon dominated tropical climate with mean annual rainfall of 3000 mm. Due to the

unique blend of the physiography, climate, and socio-economic factors, Wayanad region is highly vulnerable to the occurrence of landslides.

Methodology

Landslide inventory

A total of 263 locations where previously landslide events occurred are mapped from high resolution satellite images and verified in the field through intensive field checks. Majority of the landslides are shallow debris flow triggered by intense rainfall events.

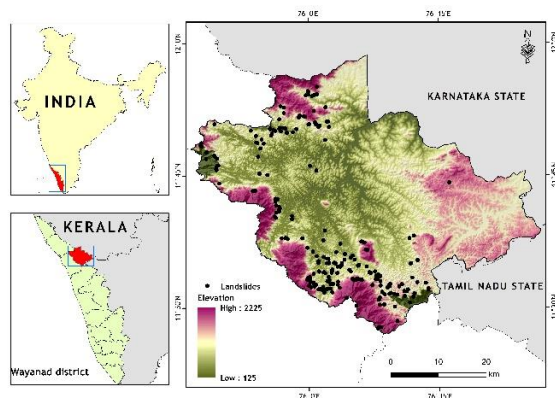


Figure 1 Location map with previous landslides

Spatial database

Landslide conditioning factors such as lithology, soil texture, land use/ land cover, slope angle, slope aspect, topographic wetness index, distance from the road, distance from the streams, distance from the lineaments, convergence index, profile and plan curvatures were used for susceptibility modelling (Achu et al., 2020). Lithology map of the study area is derived from district resources map published by Geological Survey of India. Soil texture data of the study area is obtained from Kerala State Soil Survey organization. Land use/ land cover map of the study area is gathered from Kerala State Land Use Board and stream lines were generated from SOI toposheets and Euclidian distance method is used to calculate distance from streams. Road lines of the study area is extracted from OSM data and Euclidian distance method is used to calculate distance from roads. Besides DEM derivatives such as slope angle, slope aspect, topographic wetness index, convergence index, profile and plan

curvatures were derived from SRTM DEM with 30 spatial resolution. Landsat 8 OLI images were used to derive lineaments with SRTM DEM and SOI drainage lines. All the raster files were converted to a common pixel size of 30m and ensured common projected coordinate system of WGS_1984_UTM_Zone_43N.

Deep learning neural network (DNN)

Deep learning, which is a sub-branch of ML, refers to learning processes with the use of “deep” neural network structures to infer outcomes from a set of input variables (Dou et al., 2020). Deep learning Neural Networks (DNN) is a multi-layer perceptron neural network (MLP-NN) with more than one hidden layer in its inputs and outputs are optimized by optionally Gradient descent (Kalantar et al., 2021). MLP-NN model features multiple layers that are interconnected in such a way that they form a feed-forward neural network and each neuron in one layer has directed connections to the neurons of a separate layer (Beysolow 2017). DNN are capable of handling enormous high dimensional datasets with billions of parameters that passes through non-linear functions (Kriegeskorte and Golan, 2019). However, achieving the reasonable result of a deep-learning model requires advanced hyper parameter tuning. Subsequently, in this study a preliminary analysis was carried out using grid search to find optimized hyper parameters including activation functions, hidden layers, epochs, L1 and L2 regularizations, learning rate annealing, rho, epsilon, momentum start, momentum stable and input dropout ratio. Accordingly, the one which is having maximum accuracy and least error is selected for modelling. Subsequently a DNN with 10×100×100×100×2 i.e., one input layer with 10 nodes, three hidden layers (100×100×100 nodes) with rectified linear activation function and one output layer with 2 nodes (softmax transfer function) were used for landslide susceptibility mapping. Besides the network is optimized by Stochastic gradient descent. Since the landslide susceptibility analysis is considered as a two-class classification in this study, cross entropy loss function is chosen. To avoid over fitting, training data is split for 5-fold cross validation and accuracy of each fold is estimated. Model

performance is assessed using ROC-AUC, Specificity, Sensitivity, Kappa index (K), MSE and RMSE error.

Random Forest (RF)

RF is a powerful machine-learning technique that can be applied for both classification and regression (Breinman 2001). RF builds an ensemble decision trees usually trained with the “bagging” method and merges them together to get a more accurate and stable prediction. In RF, “bagging” method selects each node of the tree from random samples of variables and observations from the training dataset for model building (Tanyu et al., 2021). Thereafter the unselected observations (out of the bag) are used to estimate the model error (OOB error), similar to standard deviation error between predicted and observed values, to establish a ranking of importance of variables (Tanyu et al., 2021). Further, to avoid less important variables from the models, a multi-collinearity test is conducted and variables with high correlation are omitted. Besides information gain (IG) based variable importance analysis was carried out to understand the predictive power of the independent variables.

Result and discussions

The probability values of landslide susceptibility index ranges from 0.00 to 9.99 which is further reclassified into five zones such as stable area (0.00-0.20) low susceptibility (0.20-0.40), moderate susceptibility (0.40-0.60), high susceptibility (0.60-0.80) and very high susceptibility (0.80-0.99) (Figure 2a and 2b). Both the models obtained reasonable accuracy in training and testing phases, the predicted DNN model shows an AUC of 0.934 in training phase and 0.887 in testing phase, whereas RF shows 0.911 and 0.850 in training and testing phases respectively. Subsequently less variation of ROC-AUC, Specificity, Sensitivity, Kappa index (K), MSE and RMSE error between the training and testing phases indicates that the mode is free from overfitting (Table 1). According to DNN method ~66.36% of the study area are mapped as stable area followed by 13.06% area under low susceptible zones, 7.73% under moderate susceptible zones, 6.92% under high susceptible

zones and 8.65% under very high susceptible zones (Figure 2a).

Table 1. Performance of the proposed DNN model in training and testing section.

	DNN		RF	
	Training	Testing	Training	Testing
Sensitivity	0.895	0.854	0.870	0.873
Specificity	0.839	0.809	0.811	0.752
K index	0.730	0.669	0.670	0.610
AUC	0.934	0.887	0.911	0.851
MSE	0.131	0.140	0.136	0.149
RMSE	0.362	0.374	0.369	0.387

Table 2. Variable importance based in IG

Parameters	average merit	SD
Slope angle	0.348	±0.003
Convergence Index	0.151	±0.003
Distance from roads	0.139	±0.004
Profile Curvature	0.114	±0.003
TWI	0.063	±0.004
LU/LC	0.060	±0.003
Plan Curvature	0.041	±0.002
Soil texture	0.029	±0.002
Lithology	0.021	±0.002
Aspect	0.011	±0.004

According IG analysis to Slope angle (AM=0.348), Convergence Index (0.151), Distance from roads (0.139) and Profile Curvature (0.114) are the major parameters which favors occurrence of landslides in the study area (Table 2). It is also noted that lithology and slope aspect show least importance in the occurrence of landslides in the study area.

Summary and Conclusion

The study utilized deep learning neural networks and machine learning random forest methods for landslide susceptibility modelling in Wayanad district of Kerala. DNN shows better predictive ability over random forest method with AUC 0.934 in training phase and 0.850 in testing phase. Therefore, DNN can be used for future hazard mitigation. According to DNN method, ~8.65% area comes under very high susceptible zones where micro level field investigations are needed

for hazard mitigation. IG based variable importance analysis shows that terrain parameters and distance from the roads are the most important variables contribute to the occurrence of landslides. Overall, the study shows that deep learning is a promising technique in landslide susceptibility analysis.

Future work: Based on the proposed landslide susceptibility map, studies needs to be initiated with precise site specific geo-technical data along

with accurate rainfall from AWS having intensity and quantity is essential for real time monitoring and early warning.

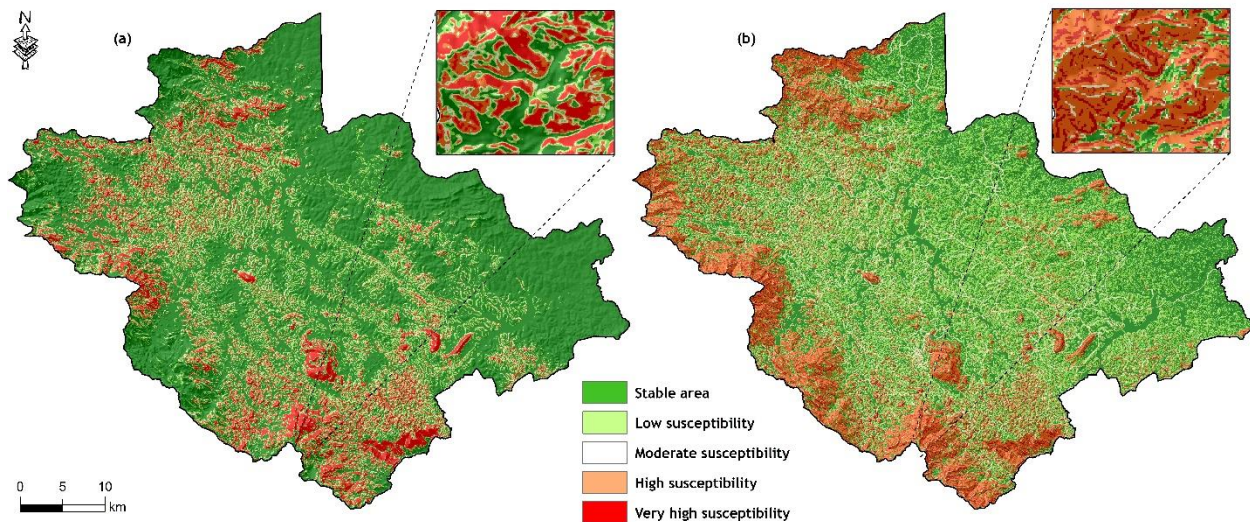


Figure 2 Landslide susceptibility map predicted by (a) DNN and (b) RF model.

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